4 (a) (i) A cubic container of sides s and cross-sectional area A contains a gas made up of a large number of molecules, each with mass m. One of these molecules has an initial velocity c and a velocity component u_x in the x direction, as shown in Fig. 4.1.

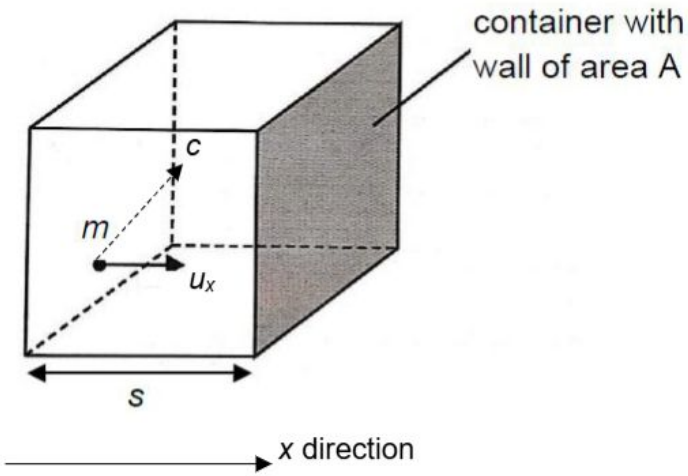


Fig. 4.1

Derive, in terms of m, u_x and s,

1. the change in momentum of the molecule when it collides with the wall on the right, [1]

Change in momentum $\Delta p = m(-u_x) - mu_x = -2mu_x$ [1]

2. the time between consecutive collisions with the wall on the right. [1]

Time between consecutive collisions on the wall, $t = \dfrac{\text{Distance travelled}}{\text{Speed}} = \dfrac{2s}{u_x}$ [1]

(ii) Hence show that the average force F exerted by the molecule on the wall on the right is given by $F = \dfrac{mu_x^2}{s}$. [1]

By Newton's Second Law, average force on particle, $F_p = \dfrac{\Delta p}{t} = \dfrac{-2mu}{2s/u} = -\dfrac{mu^2}{s}$

By Newton's Third Law, average force on wall, $F_w = -F_p = \dfrac{mu_x^2}{s}$ [1]

(iii) Each molecule of the gas has its own velocity c and velocity component u_x in the x direction.

1. State the equation that relates the mean square speed of the molecules, $<c^2>$, to the mean square speed due their velocity components in the x direction, $<u_x^2>$. [1]

As the N particles are in *random* motion, $\langle u_x^2 \rangle = \frac{1}{3}\langle c^2 \rangle$ [1]

2. Using your answers to **(a)(ii) and (a)(iii)1**, derive the expression $pV = \frac{1}{3}Nm\langle c^2 \rangle$, where p is the pressure on the wall, V is the volume of the container, N is the number of gas particles and $\langle c^2 \rangle$ is the mean square speed of the molecules. [2]

Total force of N particles on the wall, $F = N\frac{m\langle u_x^2 \rangle}{s}$ [1]

where $\langle u_x^2 \rangle$ is the mean square speed of the particles.

Pressure on the wall of the container, $p = \frac{F}{s^2} = \frac{Nm\langle u_x^2 \rangle}{s^3} = \frac{Nm\langle u_x^2 \rangle}{V}$

$$p = \frac{1}{3}\frac{Nm\langle c^2 \rangle}{V} \quad [1]$$

$$\Rightarrow pV = \frac{1}{3}Nm\langle c^2 \rangle$$

(b) A cylinder, shown in Fig. 4.2, fitted with a piston contains ideal gas at a constant temperature of 20 °C. When the volume V is 0.50×10^{-3} m^3, the pressure p in the cylinder is 20×10^4 Pa.

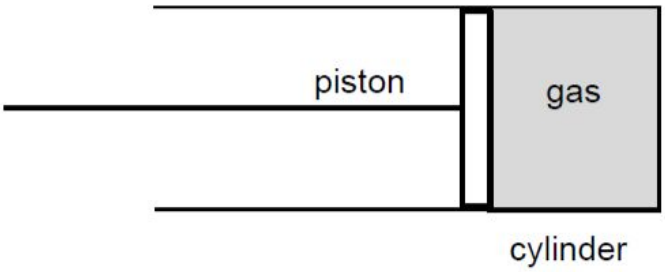


Fig. 4.2

An experiment was conducted and the data was plotted in Fig. 4.3

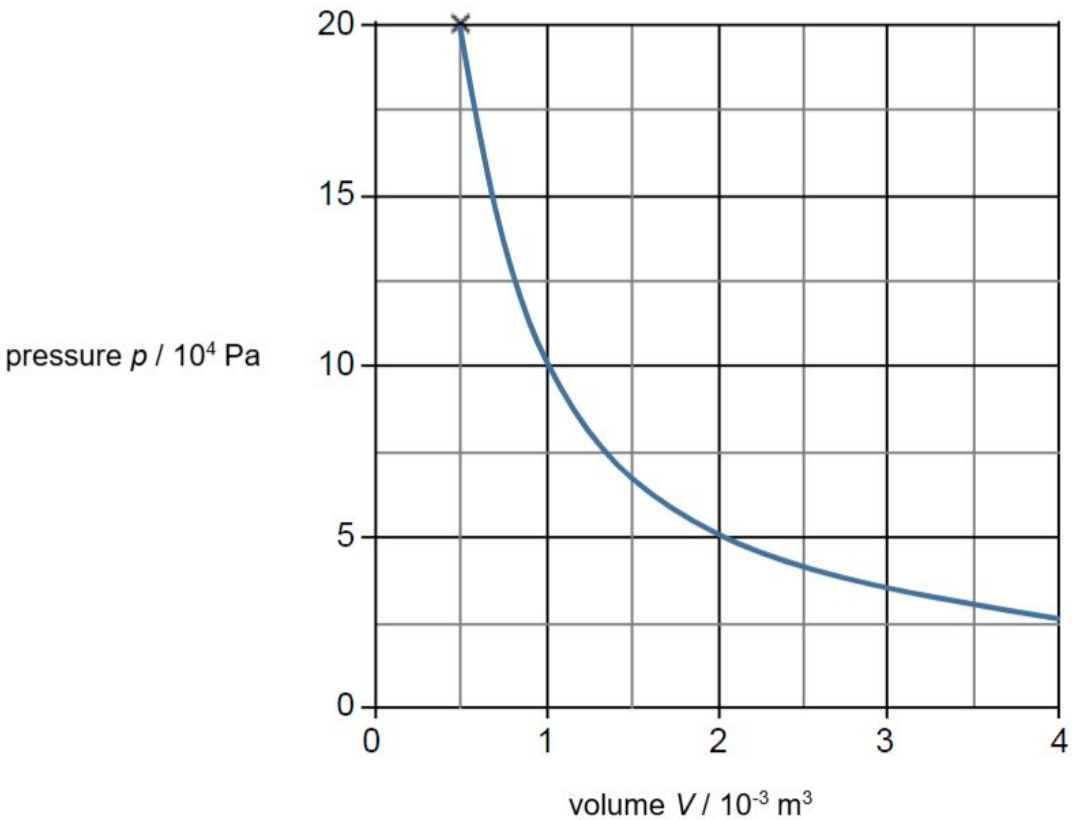


Fig. 4.3

(i) By plotting two additional points, draw a graph on Fig. 4.3 to show the relationship between p and V as the piston is slowly pulled out such that the temperature of the gas remains constant. [1]

(ii) Calculate the number of gas molecules in the cylinder.

Using the ideal gas equation, $pV = NkT$ [1]

$$\text{Number of gas molecules, } N = \frac{pV}{kT} = \frac{(20 \times 10^4)(0.5 \times 10^{-3})}{(1.38 \times 10^{-23})(20 + 273)} = 2.47 \times 10^{22} \quad [1]$$

number of gas molecules = ... [2]

(iii) Determine the total kinetic energy of the gas molecules.

Total kinetic energy, $KE_T = \frac{3}{2}NkT$

$$= \frac{3}{2}pV$$
$$= \frac{3}{2}\left(20 \times 10^4\right)\left(0.5 \times 10^{-3}\right)$$
$$= 150\,\text{J} \quad [1]$$

total kinetic energy = ... J [1]

[Total: 10]